



Electric Vehicles Analytics

Uncovering trends in EV adoption through descriptive statistics, Python EDA, and Power BI visualization.

Project Overview



Goal

Identify trends in EV usage — top manufacturers, popular models, and model year distribution.

Tools Used

Python for analysis • Power BI for interactive dashboards, KPI cards, and charts.

Dataset Summary

203

Rows

Total vehicle records

6

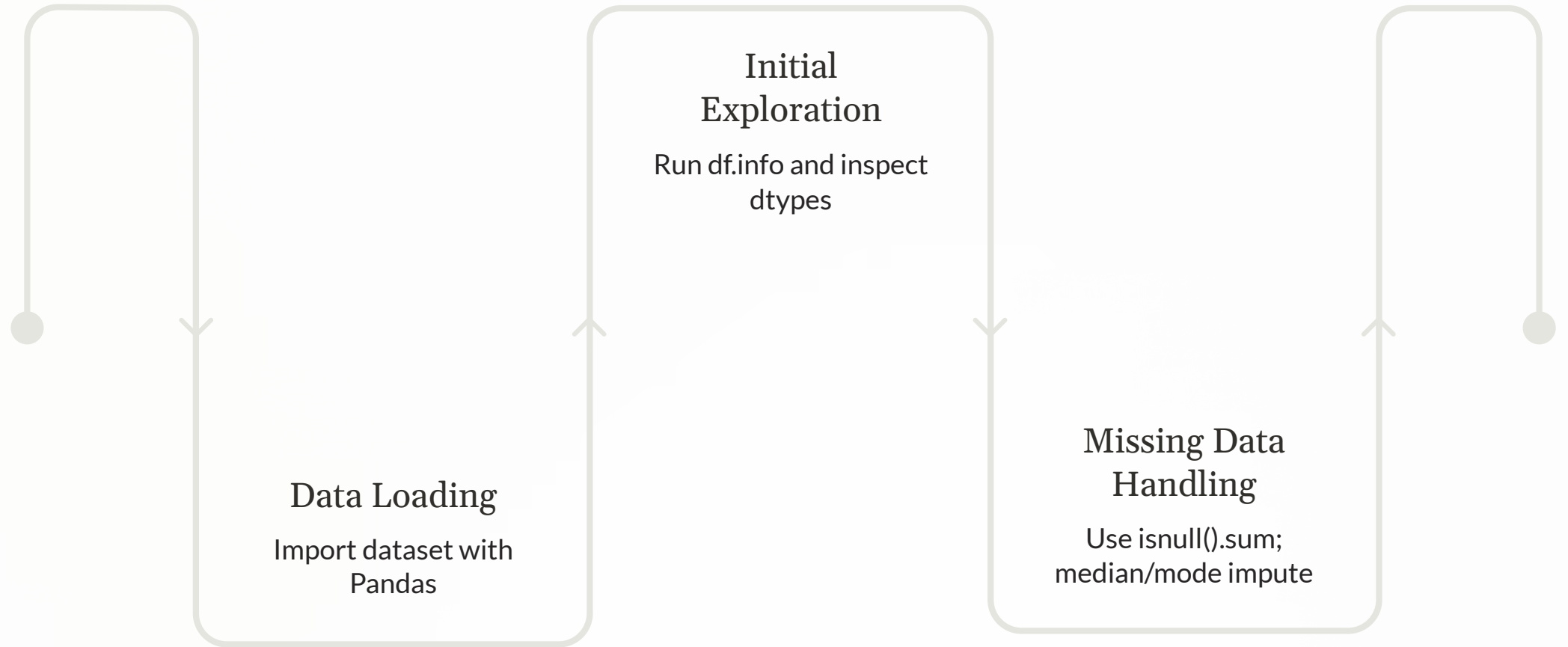
Columns

VIN, City, Model Year, Make, Model + more

- Vehicle Identification Number (object): Unique identifier of the vehicle
- City (object): The city where the vehicle is registered
- Postal code (object): The postal code associated with the vehicle's registration address
- Model year (int): The year the vehicle model was manufactured
- Make (object): Manufacturer's name

Model (object): The specific model of the vehicle

Exploratory Data Analysis



Python was used end-to-end — from loading the dataset with Pandas to cleaning and preparing data for analysis.

Data Cleaning in Python

Numerical Gaps

Model Year missing values → replaced with **median**

Categorical Gaps

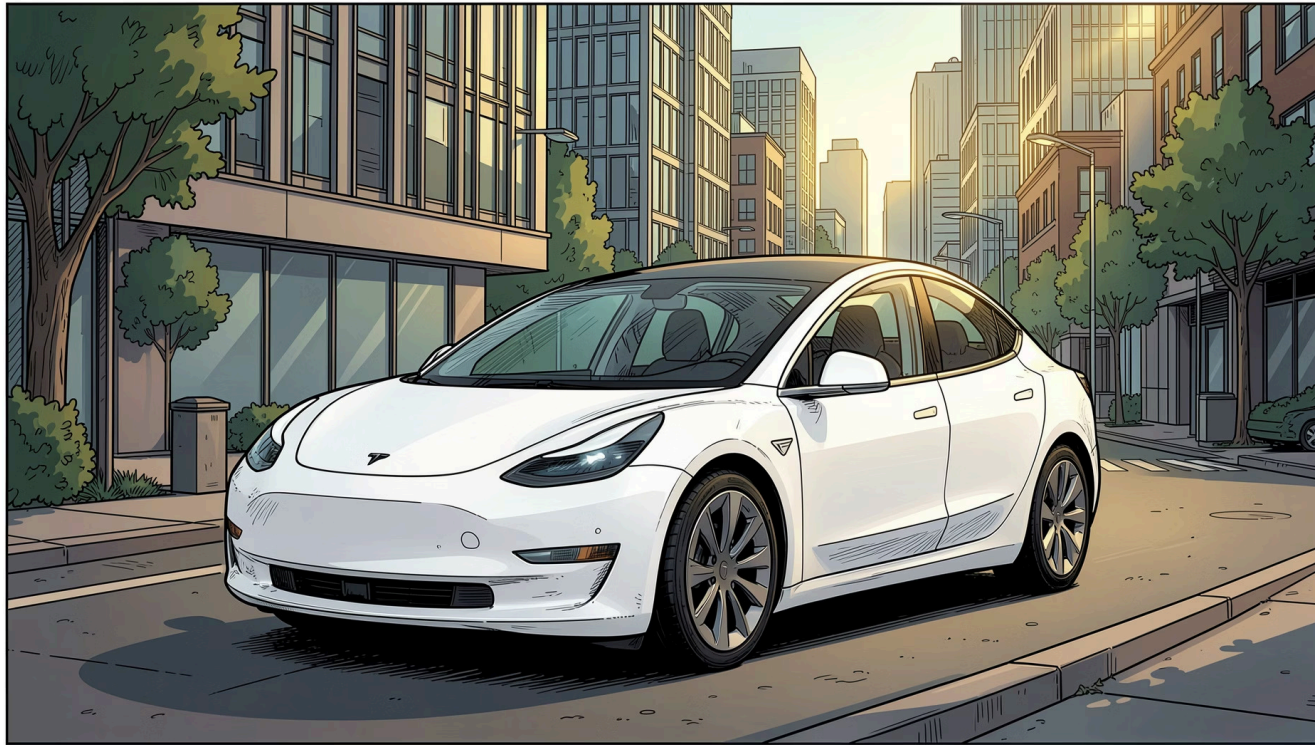
City, Make, Model, VIN → filled with **mode** or "Unknown"

Key Functions

`isnull().sum()` · `value_counts()` · `describe()`

	0
Unnamed: 0	0
vehicle_identification_number	0
city	0
postal_code	0
model_year	0
make	0
model	0

Key Insight 1: Tesla Dominates



Top Manufacturer

Tesla is the most dominant manufacturer in the dataset.

Top Model

Model 3 is the most frequently occurring vehicle.

Takeaway

Tesla's strong presence signals its central role in EV technology adoption.

Key Insight 2: Modern Fleet

Model Year Range

Most vehicles manufactured **2018 – 2025**

Average Model Year

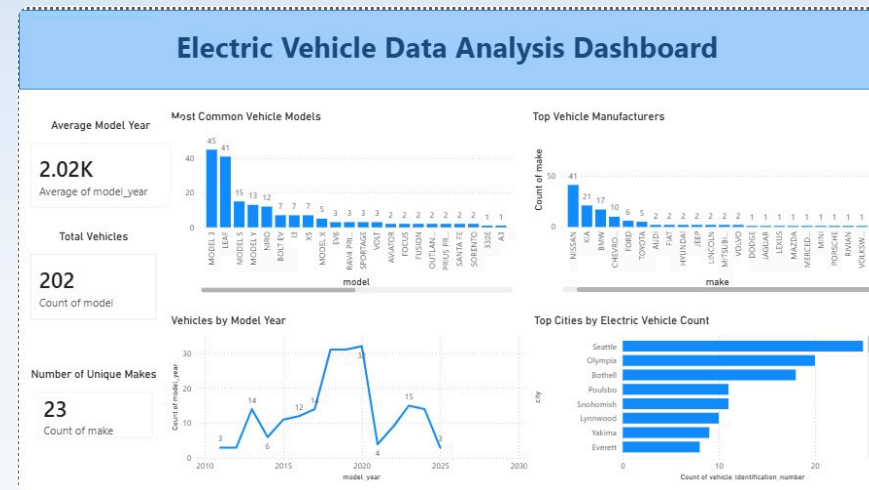
Approximately **2018.6**

Implication

Dataset reflects growing demand for **newer, advanced EVs**

Power BI Dashboard

An interactive dashboard was built to present findings visually — featuring **KPI cards**, bar charts, and line charts for clear, actionable insights.



Business Recommendations



Focus on High-Demand Models

Prioritize popular EVs like Tesla Model 3 that align with market demand.



Invest in New EV Tech

Battery efficiency, charging speed, and smart features keep companies competitive.



Expand Charging Infrastructure

Grow charging networks in urban and suburban areas to drive adoption.



Target Marketing

Use trend analysis to focus campaigns on high-adoption regions and segments.